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Nelson Mullins Riley & Scarborough LLP
IP Department
301 S. College Center, Suite 2300
301 S. College Street
Charlotte, NC 28202

EXAMINER
KIRWIN, STEFANIE JOHANNA

ART UNIT	PAPER NUMBER
1677	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ipdocket@nelsonmullins.com

Office Action Summary	Application No. 17/548,314	Applicant(s) Vander Veen et al.	
	Examiner STEFANIE J KIRWIN	Art Unit 1677	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/23/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-9 and 20-24 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9 and 20-24 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 04/25/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. The present application was filed 06/20/2024. The priority information to provisional application No. 63/124,551, filed 12/11/2020 is acknowledged.

Status of the claims

3. Claims 1-9 and 20-24 are pending and examined below. Claims 1-9 and 20-22 are amended. Claims 23 and 24 are new.

Withdrawn rejection

4. The rejection of claims 1-9 and 20-22 under 35 U.S.C. 112(b) regarding the recitation of the device comprising an aqueous extraction buffer (claim 1) and the aqueous extraction buffer (claim 22) has been withdrawn due to the amendment of the claims to a composition. See new grounds of rejection below.

Claim Rejections - 35 USC § 112

35 USC § 112(a)

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to

make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

5. **Claim 24** is rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. **This is a new matter rejection.**

New **Claim 24** recites a "competitive lateral flow assay (LFA) kit comprising: an aliquot of an aqueous extraction buffer". Applicant, at remarks page 6, recites that support for the new claims can be found throughout the specification, for example at paragraphs [0044], [0050], [0063], [0066], [0080], and [0096]-[0097], and in Figures 1 and 9. However, the specification at paragraph [0044] recites loading the test line with monensin, paragraph [0050] pertains to applying the sample to the sample pad, and paragraph [0063] recites an example comparing different extraction buffers for the extraction of monensin from feed. Paragraph [0066] describes the test strip, and paragraph [0080] recites the sample extraction protocol comprising adding the sample to the extraction buffer and paragraphs [0096]-[0097] demonstrate the reproducibility of the results with the test device. Figures 1 and 9 show a rendering of the test strip and sample application and a photograph of the test strip respectively. Further, paragraph [0049] recites combining an aliquot of animal feed with "an aliquot [...] of liquid extraction buffer, the mixture may be allowed to soak". There is no support in the specification of a kit providing an aliquot of

an aqueous extraction buffer in the context of providing the parts of the lateral flow assay and the buffer separately, but packaged together as a single invention (composition).

35 USC § 112(b)

> The following is a quotation of 35 U.S.C. 112(b):
> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.
>
> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

6. **Claims 22** and **23** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

The rejection of **claim 22** regarding the use of a trademark or trade name is maintained for the reason of record reiterated below.

Claim 22 contains the trademark/trade name Tween. Where a trademark or trade name is used in a claim as a limitation to identify or describe a particular material or product, the claim does not comply with the requirements of 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph. See *Ex parte Simpson*, 218 USPQ 1020 (Bd. App. 1982). The claim scope is uncertain since the trademark or trade name cannot be used properly to identify any particular material or product. A trademark or trade name is used to identify a source of goods, and not the goods themselves. Thus, a trademark or trade name does not identify or describe the goods associated with the trademark or trade name. In the present case, the trademark/trade name is used to identify/describe Polyethylene glycol sorbitan monolaurate and, accordingly, the identification/description is indefinite.

Claim 23 is indefinite because of the recitation of "a sample pad loaded with an animal feed sample". It is unclear whether this limitation is directed to the intended use of the claimed invention or Applicant intends to claim the device comprising a sample, and if the device does comprise a sample, how the device can be used to detect monensin in an unknown animal feed sample if a sample is already part of the device.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

7. **Claims 1-4, 6-8,** and **20-21** are rejected under 35 U.S.C. 103 as being unpatentable over **Caussette et al.** (US9182397B2), in view of **Watanabe et al.** Monoclonal-based enzyme-linked immunosorbent assay and immunochromatographic rapid assay for monensin. Analyst. 1998 Jan 1;123(12):2573-8 (see PTO-892; 12/23/2024), **Berlina et al.** Management of factors for improving antigen-antibody interaction in lateral flow immunoassay of tetracycline in human serum samples. Biomedical and Pharmacology Journal. 2019 Mar 25;12(1):17-24 (see PTO-892; 12/23/2024), **Godoy-Navajas et al.** Determination of monensin in milk samples by front-surface long-wavelength fluoroimmunoassay using nile blue-doped silica nanoparticles as labels. Talanta. 2012 May 30;94:195-200 (see PTO-892; 12/23/2024), **Janeway et al.** Immunobiology: The Immune System in Health and Disease. 5th edition. New York: Garland Science; 2001. The structure of a typical antibody molecule (see PTO-892; 12/23/2024), Li et al. CN108646036A, 10/12/2018, and Grice et al. CN107860918A, 03/30/2018.

Regarding **claim 1, and 20-21**, Caussette teaches a method for detecting an antibiotic in a sample by contacting a liquid sample with a labeled antibiotic binding protein (liquid sample; Caussette, column 2, lines 23-27) and contacting the liquid composition with the sample receiving region of a test

device (Caussette, column 2, lines 50-51), said test device configured to allow lateral flow (Caussette, column 2, line 30) and that the test device is a test strip (lateral flow assay strip; Caussette, column 5, line 41). Caussette teaches a test device and kits that can be used to detect any type of analyte in a sample, including antibiotics (kit/composition; Caussette, column 1, lines 55-58), wherein antibiotic refers to a substance that display activity against bacteria (for detection of *antibiotic*; Caussette, column 1, line 66- column 2, line 2). Caussette further teaches a test device comprising a solid support comprising a sample receiving region (sample pad), a detection region and an absorbing region (Caussette, column 3, lines 3-7). Caussette further teaches that the sample might be solid and that liquid comprising the antibiotic needs to be extracted for example by homogenization, vortexing, or solvent extraction and that the sample is preferably liquid and can be derived for example from food products including feed (Caussette, column 4, lines 7-23). Caussette further teaches that a labeled antibiotic binding protein and a labeled control reagent are part of the test device and can be present in liquid or solid form and can be located between the sample receiving region and the detection region (conjugate pad; Caussette, column 2, line 65- column 4, line 3). Caussette further teaches a detection zone comprising an immobilized antibiotic capable of binding the labeled antibiotic binding protein (immobilized *antibiotic* to which the test antibody binds) and a control zone comprising an immobilized binding agent capable of binding the labeled control reagent (*agent* to which the control antibody binds; *Caussette*, column 2, lines 23-24). Caussette further teaches that the zones can be in a variety of configurations including lines (test/control line; Caussette, column 7, lines 63-34). Caussette further teaches labeling the antibiotic binding protein and the control reagent with gold particles (detectable label; Caussette, column11, lines 30-32). Caussette further teaches that the binding proteins can be an antibody (specific antibody; Caussette, column 5, lines 1-4) and that the control reagent is IgY (control antibody; Caussette, column 11, line 46). Caussette further teaches that anti-IgY antibody was applied to the control zone (immobilized *binding partner* to which C antibody binds; Caussette, column 11, lines 7-

9) and that the labeled control reagent is IgY-coated gold particles (Caussette, column 11, lines 46-47). Caussette further teaches that the control zone may comprise at least one capture reagent and that the capture reagent refers to any reagent that can be used to create the required functionality in the control zone (Caussette, column 8, lines 4-11). Caussette further teaches that the preferred embodiment of the capture reagent at the control zone is a member of a binding pair, e.g. a specific binding pair such as an antigen/antibody pair (Caussette, column 8, lines 37-40). As such, although the exemplary capture reagent in the control zone of Caussette is an anti-IgY antibody, Caussette also teaches a capture agent immobilized in the control zone that is an antigen (immobilized antigen to which the C antibody binds). Caussette further teaches a test device having a proximal and a distal end, said test device configured to allow lateral flow from the proximal to the distal end comprising in sequence from the proximal to the distal end a sample receiving region, a detection region and an absorbing region (liquid sample flows through the conjugate pad and then across the test line and control line; Caussette, column 2, lines 29-47). Caussette further teaches that the detection zone and control zone were applied onto a nitrocellulose membrane (a membrane surface comprising a test line and a control line; Caussette, column 11, lines 14). Caussette further teaches that the detection zone and the control zone each may comprise at least one capture reagent (Caussette, column 8, lines 4-5) and that the zones are capable of generating a signal, for instance a visual color signal, upon presence or absence of the complex between the capture reagent and its binding partner (binding of antibody results in a detectable signal; Caussette, column 8, lines 14-17). Caussette further teaches that when the label density in the detection zone is higher than that in the control zone, the sample contains no antibiotic or antibiotic at a concentration below a given threshold and when the label density in the detection zone is less than the label density in the control zone, antibiotic is present in the sample at a concentration above a given threshold (presence of antibiotic in the liquid sample decreases the detectable signal relative to a sample lacking antibiotic; Caussette, column 5, lines 46-56). Caussette further teaches that the labeled antibiotic

binding protein and the labeled control reagent are part of the test device (Caussette, column 3, lines 65-66). Caussette further teaches that compounds facilitating the resuspension and/or dissolution and/or mixing of the labeled antibiotic binding protein, the labeled control reagent, and the sample can be added and that suitable compounds include a buffer such as a Tris buffer (Caussette, column 3, lines 27-36). Put another way, Caussette, in teaching a compound facilitating the dissolution of the sample, teaches an extraction buffer, such as Tris buffer, that is an aqueous buffer.

Caussette fails to teach that the antibiotic is monensin. Caussette further fails to teach 30-40 µl (claim 1) or specifically 30 µl (claim 20) or 40 µl (claim 21) of anti-monensin specific antibody and that the device has a sensitivity of 10 ppm or less.

Watanabe teaches a monoclonal-based an immunochromatographic rapid assay (comprising direct detection) for monensin (Watanabe, see title). Watanabe teaches a rapid test kit based on the immunochromatographic method to detect monensin in cattle milk, cattle plasma, and chicken plasma (Watanabe, page 2573, Abstract, lines 8-10 and Figure 4). Watanabe further teaches diluting the sample in an assay buffer that is Tris-HCl and 0.05% Polyethylene glycol sorbitan monolaurate-20 (Watanabe, page 2574, 'Buffers', lines 1-3).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette, in order to have provided anti-monensin binding protein as the antibiotic binding protein located in the conjugate pad and further modify the test line with an immobilized binding reagent capable of binding the monensin binding protein, as an obvious matter of a simple substitution of one art recognized antibiotic binding protein (namely a monensin binding protein) and binding reagent (namely monensin) over another. The prior art contained the base invention (see as taught by Caussette, the prior art already recognized a rapid diagnostic test strip configured for a competitive assay for detecting antibiotics in a liquid sample, such as a liquid sample derived from animal feed). Further, at the time the prior art recognized the

ability of detecting monensin in a lateral flow assay (as in Watanabe). As such, it would have been obvious to modify the test strip of Caussette to capture monensin, as the ordinarily skilled artisan would appreciate one binding protein is usable in place of the other, given that Caussette teaches a lateral flow device for a competitive antibiotic detection assay and Watanabe teaches detection of monensin by a lateral flow device, and one of ordinary skill in the art would expect success either in a competitive (as taught by Caussette) or direct (as taught by Watanabe) assay, which are merely two different formats of a lateral flow assay, both doable on a single lateral flow device. The results would have been predictable, namely in that the device of Caussette in view of Watanabe would be expected capable of detecting monensin. The teaching of Watanabe further supports an expectation of success in using an aqueous buffer for the detection of monensin using a lateral flow assay because Watanabe teaches success detecting monensin in a liquid sample diluted in an aqueous buffer (namely Tris-HCl buffer).

Berlina teaches a lateral flow immunoassay for the detection of antibiotics (Berlina, page 17, Abstract, line 3). Berlina further teaches that in a competitive lateral flow assay the concentration of the specific antibody is one of the most important parameters and that increasing antibody concentration over a concentration that gives an adequate analytical signal increases unspecific staining (Berlina, page 20, 2nd column, see entire 2nd paragraph).

Godoy-Navajas teaches a competitive heterogeneous immunoassay with antibody capture involving immobilization of BSA-monensin. Godoy-Navajas further teaches adding 75 µl of 10 nM NP-labeled anti-monensin antibody (Godoy-Navajas, page 197, see '2.3.4. Determination of monensin'). Considering that an IgG antibody is defined by the state of the art as having a molecular weight of approximately 150 kDa (see e.g., Janeway, see 3-1., lines 1-2), Godoy-Navajas teaches adding 20 µg/ml labeled antibody.

Li et al. teaches a lateral flow device (see Li et al. CN108646036A, original, page 1 of 10). Li et al. further teaches a test strip to detect follicular stimulating hormone and interstitial cell stimulating

hormone comprising a bonding pad comprising mouse anti-alpha follicular stimulating hormone monoclonal antibodies compromising colloid gold label (Li, CN108646036A, translation, from here on out referred to as Li, page 1 of 6, 'Abstract', lines 1-4). Li further teaches that the gold-labeled antibody crosses the nitrocellulose membrane (Li, page 4 of 6, paragraph 11, lines 1-2). Li further teaches the strip comprising 25-35 μg/ml gold labelled mouse anti-alpha-FSH monoclonal antibody on the bonding pad (Li, page 1 of 6, see paragraph '2.'). Li further teaches an embodiment of the test strip comprising 30 μg/ml gold labeled antibody (Li, page 4 of 6, 'Embodiment one', see entire 3rd paragraph). Put another way, Li teaches 25-35 μg/ml gold labeled antibody loaded on the conjugate pad capable of binding a specific analyte (equivalent to the "M antibody").

Grice teaches a colloidal gold immunochromatographic test strip comprising detection lines, antigen lines and quality control lines and an anti-target protein monoclonal antibody-colloidal gold marker on the gold label pad for detecting gluten (Grice, page 1 of 6, 'Abstract', lines 1-8). Grice further teaches that the concentration of the anti-gluten protein monoclonal antibody-colloidal gold marker is 2-30 μg/ml (Grice, page 2 of 6, 'Contents of the invention', see 7th paragraph). Grice further teaches that when the sample is added to the sample pad it migrates to the gold label pad and if there is analyte to be detected it will combine with the gold-labeled antibody to form a complex and binds to the capture protein on the detection line (Grice, page 2 of 6, see entire last paragraph). As such, Grice also teaches a gold labeled antibody capable of binding an analyte in a sample wherein the bound analyte/antibody complex migrate to the detection line (equivalent to the "M antibody") and the antibody is at a concentration of 2-30 μg/ml.

Regarding the limitations 30-40 μg/ml (claim 1) and 30 μg/ml (claim 20) and 40 μg/ml (claim 21) as presently recited in the claims; it has been previously established by the courts that a *prima facie* case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). In the present case, 30-40 μg/ml, 30 μg/ml, and 40 μg/ml is an

amount overlapping with or substantially close to what was taught in the prior art. Both Li and Grice teach a colloidal gold labeled antibody that binds an analyte in a sample and migrates along the lateral flow test strip, such as 25-35 µg/ml gold labelled antibody as taught by Li, 2-30 µg/ml gold labeled antibody as taught by Grice, or 20 µg/ml as taught by Godoy-Navajas. Berlina teaches that the concentration of the specific antibody is one of the most important parameters. Berlina further teaches providing enough antibody to give an adequate analytical signal and that increasing antibody concentration increases unspecific staining (Berlina, page 20, 2nd column, see entire 2nd paragraph). As such it would have been considered *prima facie* obvious in view of the prior art to have arrived at the presently claimed antibody concentration in the course of routine experimentation, in order to achieve a desired effect of detecting monensin in a sample with an adequate signal, but without an increase in non-specific binding.

The courts have also held that a particular parameter, when recognized as a result effective variable, i.e., a variable that achieves a recognized result, may be subject to routine optimization in order to determine optimum or workable ranges of said variable (also MPEP 2144.05). In the present case, it would have been *prima facie* obvious to one of ordinary skill in the art to determine the optimum antibody concentration by optimizing what was known in the art to be a suitable concentration of antibodies as taught by Godoy-Navajas, Li, and Grice. Godoy-Navajas, Li, and Grice show that it was known in the art to use an antibody concentration of 2-35 µg/ml gold labeled mobile detection antibody for a lateral flow immunoassay. It would have been *prima facie* obvious to have arrived at the claimed range between 30 and 40 µg/ml as an obvious matter of routine experimentation because the antibody concentration is an art recognized result effective variable (namely the variable which results in successful detection of monensin, that it would be desirable to optimize conditions to uncover the optimal concentration in order to avoid unnecessary increase in non-specific signal, see Berlina, concentration of the specific antibody is one of the most important parameters and comprising

enough antibody to give an adequate signal and not so much that it increases unspecific staining). It would have further been obvious to have modified the 2-35 µg/ml in order to arrive at the claimed range/concentration values, one motivated to optimize the 2-35 µg/ml be selecting from and trying values within this prior art disclosed range because of the teaching of Berlina that the concentration of the specific antibody is one of the most important parameters in a competitive lateral flow assay.

One of ordinary skill in the art would have a reasonable expectation of success because an antibody concentration of 2-35 µg/ml is substantially similar, and overlaps with the lower concentration of the range 30-40 µg/ml, and considering the values are substantially close, there is no reason to expect that the values within the claimed range would achieve a different or unexpected result compared to 2-35 µg/ml. One of ordinary skill would expect success optimizing by routine experimentation because such optimization would allow the operator to achieve the desired detection of monensin and further because optimization is well within the skill level of the ordinary artisan, as supported by the above cited prior art.

Regarding the limitations that the LFA strip device has a sensitivity of 10 ppm or less for detecting antibiotic in the liquid sample, as indicated in the citation of Caussette and the combined cited art above, the claimed invention is structurally indistinct from the cited prior art, and as such, would be expected capable of the same sensitivity.

Regarding **claims 2** and **3**, Caussette and the cited art above teach a composition substantially as claimed.

With respect to the recitations “which has a sensitivity of 2 ppm” (claim 2) or “which has a sensitivity or 6-10 ppm for detecting monensin in the liquid sample” (claim 3), applicant is reminded that a limitation of the claimed invention must result in a structural difference between the claimed invention and the prior art in order to patentably distinguish the claimed invention from the prior art. If

the prior art structure is capable of achieving the limitation, then it meets the claim. Applicant is also reminded that claim scope is not limited by claim language that does not limit a claim to a particular structure. See MPEP 2111.04. In this case, no clear structural differences are invoked by the recitation of the sensitivity as claimed. Because there is no distinction between the claimed composition and that of the cited prior art, it is the case that the composition of the lateral flow device and the buffer as taught by Caussette and the prior art above would similarly be expected capable of achieving the same sensitivity. The limitations of claims 2 and 3 fail to clearly require any additional reagent of component be part of the composition and fail to impart any particular structural feature or limitation.

Regarding **claims 4** and **6**, Caussette teaches that the sample receiving section is made of a porous material (Caussette, column 7, lines 29-30). Caussette further teaches that porous material refers to any material capable of providing lateral flow, for example polymeric materials such as polyester (claim 4), also cotton, glass fiber, nitrocellulose and blends of nitrocellulose with polymeric materials (claim 6; Caussette, column 9, lines 31-35).

Regarding **claim 7**, Caussette teaches labeling the antibiotic binding protein with colloidal gold particles (Caussette, column 5, lines 27-28).

Regarding **claim 8**, Caussette teaches covalently conjugating the capture reagent to the detection region through a carrier such as bovine serum albumin (BSA), optionally coupling the capture reagent to the carrier by a spacer (Caussette, column 8, lines 49-55).

8. **Claim 5** is rejected under 35 U.S.C. 103 as being unpatentable over **Caussette et al.**, in view of **Watanabe et al.**, **Berlina et al.**, **Godoy-Navajas et al.**, **Janeway et al.**, **Li et al., and Grice et al.** as

applied to claim 1 above, and further in view of **Salminen et al.** Ultrasensitive and robust point-of-care immunoassay for the detection of Plasmodium falciparum malaria. Analytical Chemistry. 2020 Nov 23;92(24):15766-72 (see PTO-892; 12/23/2024).

Regarding **claim 5**, Caussette and the cited art above teach a competitive lateral flow device for the detection of monensin substantially as claimed.

Caussette does not teach that the conjugate pad is a chopped glass pad.

Salminen teaches an optimized highly sensitive lateral flow device (Salminen, 'Abstract', lines 6-8). Salminen further teaches testing 10 different conjugate pads and choosing the Ahlstrom 8950 glass fiber conjugate pad (Salminen, page 15768, 3rd paragraph, lines 14-18). Salminen further teaches that Ahlstrom 8950 is made of chopped glass (Salminen, see supplemental material, page S-3, supplemental Table S2) and has the best signal/background ratio of the materials tested (Salminen, see supplemental material, page S-7, figure S5).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette with the chopped glass conjugate pad of Salminen because of the teaching of Salminen that the chopped glass conjugate pad resulted in the best signal/background ratio out of 10 tested conjugate pad materials.

One having ordinary skill in the art would have a reasonable expectation of success because Salminen teaches success using the chopped glass conjugate pad in a lateral flow device, the same device as taught by Caussette.

9. **Claim 9** is rejected under 35 U.S.C. 103 as being unpatentable over **Caussette et al.**, in view of **Watanabe et al.**, **Berlina et al.**, **Godoy-Navajas et al.**, **Janeway et al.**, **Li et al., and Grice et al.** as applied to claim 1 above, and further in view of **Bisgrove et al.**, US9945850B2.

Regarding **claim 9**, Caussette and the cited art above teach a competitive lateral flow composition substantially as claimed.

Caussette teaches that the control reagent is IgY-coated gold particles (Caussette, column 11, line 46-47) and that anti-IgY antibody was applied to the control zone (Caussette, column 11, lines 7-9) and that the labeled control reagent is IgY-coated gold particles (Caussette, column 11, lines 46-47).

Caussette does not teach that the control antibody is gold-labeled anti-chicken IgY or anti-mouse IgG antibody and that the immobilized antigen at the control line is chicken IgY or mouse IgY antibody.

Bisgrove teaches a lateral flow assay test strip (Bisgrove, column 5, lines 57-58). Bisgrove further teaches a control region containing immobilized control agents which specifically bind to mobile control binding agents to form a control binding pair (Bisgrove, column 9, lines 2-5). Bisgrove further teaches that examples of suitable control binding pairs of interest include chicken IgY/anti-chicken IgY and that either member of these pairs may be the immobilized control agent with the other being the control binding agent (Bisgrove, column 9, lines 12-17). Bisgrove further teaches that control binding pairs of interest act as internal controls against which the analyte measurement results may be compared on the individual test strip.

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette to have used a gold labeled anti-IgY antibody and to have immobilized the IgY antibody to the control line as an obvious matter of a simple substitution of one art recognized way of immobilizing one partner of a control binding pair and labeling the other one over another, both recognized as suitable for the same purpose, both acting as a suitable control. The ordinarily skilled artisan would be motivated to do so because labeling the IgY-antibody and immobilizing an anti-IgY antibody in the control zone performs the same

function specified in the claim in substantially the same way and produces substantially the same results of acting as an internal control against which the analyte measurement results may be compared on the individual test strip.

One of ordinary skill in the art would have a reasonable expectation of success in doing so because both Caussette and Bisgrove teach a lateral flow assay comprising IgY and anti-IgY antibody as the control binding pair.

10. **Claim 22** is rejected under 35 U.S.C. 103 as being unpatentable over **Caussette et al.**, in view of **Watanabe et al.**, **Berlina et al.**, **Godoy-Navajas et al.**, **Janeway et al.**, **Li et al., and Grice et al.** as applied to claim 1 above, and further in view of **Mount et al.** Production of antibodies and development of enzyme immunoassay for determination of monensin in biological samples. Journal of the Association of Official Analytical Chemists. 1987 Mar 1;70(2):201-5, **Rudolf et al.** Effects of different extraction buffers on peanut protein detectability and lateral flow device (LFD) performance. Food Additives & Contaminants: Part A. 2012 Jan 1;29(1):1-1, and **Granados-Chinchilla F.** A focus on aflatoxin in feedstuffs: new developments in analysis and detection, feed composition affecting toxin contamination, and interdisciplinary approaches to mitigate it. Aflatoxin-Control, Analysis, Detection and Health Risks. 2017 Aug 30:251-80.

Regarding **claim 22**, Caussette and the cited art above teach a competitive lateral flow composition substantially as claimed.

Caussette and the cited art above does not teach that the aqueous extraction buffer comprises phosphate buffered saline and 1% Polyethylene glycol sorbitan monolaurate.

Mount teaches an enzyme immunoassay for determination of monensin in biological samples (Mount, see title). Mount further teaches that the samples and monensin standards are diluted in

phosphate buffered saline Polyethylene glycol sorbitan monolaurate (Mount, page 201, Figure 2, legend line 6). Mount further teaches that the phosphate-buffered saline is PBS containing 0.05% Polyethylene glycol sorbitan monolaurate-20 (Mount, page 202, see '(g)').

Rudolf teaches comparing and optimizing extraction buffers and running conditions for the detection of peanut protein using immunochromatographic test strips (Rudolf, page 2, 2nd paragraph, lines 1-4). Rudolf teaches comparing Phosphate, carbonate, Tris/HCl, Citrate, Borate, and Phosphate/citrate buffers with various additives, comprising no additive, 1 M NaCl, 1% Tween (Polyethylene glycol sorbitan monolaurate), 1% SDS, and 6 M Urea (Rudolf, page 2, see table 1). Rudolf further teaches comparing extraction from in-shell-roasted peanuts (easily extracted) and dark-roasted peanuts (very poor extractability). Rudolf further teaches that because of the difference in extractability samples from which the proteins where extracted easily (i.e. in-shell-roasted peanuts) and those sample types that showed poor extractability of proteins (dark-roasted peanuts) where used to compare extraction buffers (Rudolf, page 3, 'Results', line 1-page 4, line 6). Rudolf further teaches that buffers containing Tween-20 might be more adequate for application in lateral flow assays as they are more unlikely to interfere with matrix substances of some products (Rudolf, page 9, lines 2-5). Rudolf teaches that citrate buffer and all buffers containing SDS or urea did not show satisfying protein extraction and lateral flow compatibility. Buffers that gave good results in extraction or lateral flow compatibility could not compensate for an insufficient performance in one step by being well suited for the other (extraction step vs. lateral flow compatibility) and that carbonate and borate buffer, which showed good extraction capacities are not compatible with lateral flow requirements. Finally, Rudolf teaches, the best results were obtained with phosphate buffer without additives and with 1% Tween-20 (Polyethylene glycol sorbitan monolaurate-20, Rudolf, page 8, see entire second paragraph 'Total buffer performance'). Put another way, Rudolf teaches the importance of the extraction buffer in samples that

have poor extractability due to interference by the sample matrix, for example in some food stuff, in order to achieve good extraction while maintaining compatibility with the lateral flow assay.

Granados-Chinchilla teaches that animal feed is a complex matrix (Granados-Chinchilla, Abstract, lines 7-8). Granados-Chinchilla further teaches that in complex samples high matrix interference represent a limitation (Granados-Chinchilla, page 206, 4th paragraph, lines 1-3).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the composition of Caussette, namely the extraction buffer of Caussette, in order to use phosphate buffered saline comprising 1% Polyethylene glycol sorbitan monolaurate-20, as taught by Rudolf, in place of the Tris buffer of Caussette because of the teaching of Rudolf that best results for extracting a sample and applying it to a lateral flow assay where achieved with phosphate buffered saline and that buffers containing Tween-20 might be more adequate for application in lateral flow assays as they are more unlikely to interfere with matrix substances. One of ordinary skill in the art would have been motivated to do so because of the teaching of Granados-Chinchilla that animal feed is a complex matrix and that in complex samples matrix interference represents a limitation and the success of Rudolf using phosphate buffered saline comprising 1% Polyethylene glycol sorbitan monolaurate-20 to extract protein from such a complex matrix while being compatible with a lateral flow assay.

One having ordinary skill in the art would have had a reasonable expectation of success in doing so because of the teaching of Mount that demonstrates success of detecting monensin in a sample diluted in phosphate buffered saline comprising 0.05% Polyethylene glycol sorbitan monolaurate-20 by immunoassay and the teaching of success by Rudolf of phosphate buffered saline comprising 1% Polyethylene glycol sorbitan monolaurate-20 for both extraction of protein and the lateral flow assay. As a result, one would have a reasonable expectation of success, substituting the extraction buffer as

taught by Caussette with another buffer known to be suitable for protein extraction from a complex matrix and a lateral flow assay and used in immunoassay detection of monensin.

Regarding the limitation 1% Polyethylene glycol sorbitan monolaurate-20 as presently recited in the claims; it has been previously established by the courts that a *prima facie* case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). In the present case, 1% Polyethylene glycol sorbitan monolaurate-20 is an amount that is taught by Rudolf in the extraction of peanut protein and substantially close to 0.05% as taught by Mount for the detection of monensin and as such it would have been *prima facie* obvious in view of Mount to have arrived at the presently claimed Polyethylene glycol sorbitan monolaurate-20 concentration for extraction of monensin out of the course of routine experimentation, in order to achieve a desired effect of extracting monensin from a complex matrix and suspending a sample comprising monensin in a liquid for application on a lateral flow device.

The courts have also held that a particular parameter, when recognized as a result effective variable, i.e., a variable that achieves a recognized result, may be subject to routine optimization in order to determine optimum or workable ranges of said variable (also MPEP 2144.05). In the present case it would have been prima facie obvious to one of ordinary skill in the art to determine the optimum Polyethylene glycol sorbitan monolaurate -20 concentration by optimizing what was known in the art to be a suitable concentration of Polyethylene glycol sorbitan monolaurate-20 to dilute monensin as taught by Mount. Mount teaches that it was known in the art to use a Polyethylene glycol sorbitan monolaurate-20 concentration of 0.05% for the dilution of monensin and Rudolf teaches the suitability of phosphate buffered saline comprising 1% Polyethylene glycol sorbitan monolaurate-20 for protein extraction from a complex matrix and for further use in a lateral flow assay. It would have been obvious to have arrived at the claimed concentration of 1% Polyethylene glycol sorbitan monolaurate-20 for the extraction and dilution of monensin as an obvious matter of routine experimentation because the

Polyethylene glycol sorbitan monolaurate-20 concentration is an art recognized result effective variable (namely the variable which results in successful detection of monensin).

One of ordinary skill in the art would have a reasonable expectation of success because Rudolf teaches success using PBS buffer comprising 1% Polyethylene glycol sorbitan monolaurate-20 to extract protein from a complex food matrix (dark roasted peanut) and using it as buffer in a lateral flow device and Mount teaches success using PBS buffer comprising 0.5% Polyethylene glycol sorbitan monolaurate-20 in an immunoassay for the detection of monensin. Polyethylene glycol sorbitan monolaurate-20 concentration of 0.05% is substantially similar to 1%. One of ordinary skill would expect success optimizing by routine experimentation because such optimization would allow the operator to achieve the desired detection of monensin and further because optimization is well within the skill level of the ordinary artisan.

11. **Claims 23-24** are rejected under 35 U.S.C. 103 as being unpatentable over **Caussette et al.**, in view of **Watanabe et al.**, **Berlina et al.**, **Godoy-Navajas et al.**, **Janeway et al., Li et al., and Grice et al.** and further in view of Matsuoka et al. Review of monensin toxicosis in horses. Journal of Equine Veterinary Science. 1996 Jan 1;16(1):8-15.

Regarding **claims 23** and **24**, Caussette teaches a method, test device and kit for detecting antibiotics in a sample (Caussette, column 1, lines 55-65). Caussette further teaches that said test device is configured to allow lateral flow (Caussette, column 2, line 30) and that the test device is a test strip (lateral flow assay strip device; Caussette, column 5, line 41). Caussette teaches that antibiotics might be present in food products including feed (Caussette, column 4, lines 18-23). Caussette teaches a method for detecting an antibiotic in a sample comprising contacting a liquid sample with a labeled antibiotic binding protein (Caussette, column 2, lines 23-27) and contacting the liquid composition with the

sample receiving region of a test device (Caussette, column 2, lines 50-51). Caussette further teaches a test device comprising a solid support comprising a sample receiving region (sample pad), a detection region and an absorbing region (Caussette, column 3, lines 3-7). Caussette further teaches that the sample might be solid and that liquid comprising the antibiotic needs to be extracted for example by homogenization, vortexing, or solvent extraction and that the sample is preferably liquid and can be derived for example from food products including feed (Caussette, column 4, lines 7-23). Caussette further teaches that a labeled antibiotic binding protein and a labeled control reagent are part of the test device and can be present in liquid or solid form and can be located between the sample receiving region and the detection region (conjugate pad). Caussette further teaches a detection zone comprising an immobilized antibiotic test line capable of binding the labeled antibiotic binding protein and a control zone comprising an immobilized binding agent capable of binding the labeled control reagent (control line; Caussette, column 2, lines 23-24). Caussette further teaches that the zones can be in a variety of configurations including lines (test/control line; Caussette, column 7, lines 63-34). Caussette further teaches labeling the antibiotic binding protein and the control reagent (Caussette, column11, lines 30-31). Caussette further teaches that the binding proteins can be an antibody (Caussette, column 5, lines 1-4). Caussette further teaches that anti-IgY antibody was applied to the control zone (immobilized *binding partner* to which C antibody binds; Caussette, column 11, lines 7-9) and that the labeled control reagent is IgY-coated gold particles (Caussette, column 11, lines 46-47). Caussette further teaches that the control zone may comprise at least one capture reagent and that the capture reagent refers to any reagent that can be used to create the required functionality in the control zone (Caussette, column 8, lines 4-11). Caussette further teaches that the preferred embodiment of the capture reagent at the control zone is a member of a binding pair, e.g. a specific binding pair such as an antigen/antibody pair (Caussette, column 8, lines 37-40). As such, although the exemplary capture reagent in the control zone of Caussette is an anti-IgY antibody, Caussette also teaches a capture agent immobilized in the control

zone that is an antigen (immobilized antigen to which the C antibody binds). Caussette further teaches a test device having a proximal and a distal end, said test device configured to allow lateral flow from the proximal to the distal end comprising in sequence from the proximal to the distal end a sample receiving region, a detection region and an absorbing region (Caussette, column 2, lines 29-47). Caussette further teaches that the detection zone and control zone were applied onto a nitrocellulose membrane (a membrane surface comprising a test line and a control line) and that the membrane was blocked by spraying a solution comprising Tris buffer (device comprises an aqueous buffer; Caussette, column 11, lines 14). Caussette further teaches that the detection zone and the control zone each may comprise at least one capture reagent (Caussette, column 8, lines 4-5) and that the zones are capable of generating a signal, for instance a visual color signal, upon presence or absence of the complex between the capture reagent and its binding partner (binding of antibody results in a detectable signal; Caussette, column 8, lines 14-17). Caussette further teaches that when the label density in the detection zone is higher than that in the control zone, the sample contains no antibiotic or antibiotic at a concentration below a given threshold and when the label density in the detection zone is less than the label density in the control zone, antibiotic is present in the sample at a concentration above a given threshold (presence of antibiotic in the liquid sample decreases the detectable signal relative to a sample lacking antibiotic; Caussette, column 5, lines 46-56). Caussette further teaches that the labeled antibiotic binding protein and the labeled control reagent are part of the test device (Caussette, column 3, lines 65-66). Caussette further teaches that compounds facilitating the resuspension and/or dissolution and/or mixing of the labeled antibiotic binding protein, the labeled control reagent, and the sample can be added and that suitable compounds include a buffer such as a Tris buffer (Caussette, column 3, lines 27-36). Put another way, Caussette, in teaching a compound facilitating the dissolution of the sample, teaches an extraction buffer, such as Tris buffer, that is an aqueous buffer.

Caussette teaches a kit comprising a labeled antibiotic binding protein, a labeled control reagent, and a test device according to the present invention (Caussette, column 10, lines 4-6) and further teaches that compounds facilitating resuspension and/or dissolution and/or mixing of the labeled antibiotic binding protein, the labeled control reagent and the sample which include buffers such as a Tris buffer, a surfactant such as Triton X-100, and other compounds (Caussette, column 3, lines 27-37) and that these compounds are present in a container and that the labeled antibiotic binding protein and the labeled control reagent may be present in different containers (Caussette, column 3, lines 51-53). Alternatively, Caussette teaches that a labeled antibiotic binding protein and a labeled control reagent are part of the test device and can be present in liquid or solid form and can be located between the sample receiving region and the detection region (conjugate pad). Put another way, Caussette teaches a kit comprising the test device comprising a sample pad, a conjugate pad loaded with labeled antibiotic specific antibody and labeled control antibody, a membrane comprising a test line and a control line and a container comprising compounds such as of an aqueous extraction buffer.

Caussette fails to teach that the antibiotic is monensin. Caussette further fails to teach a sample pad loaded with an animal feed sample. Caussette further fails to teach 30-40 µg anti-monensin specific antibody and that the LFA strip device has a sensitivity of 10 ppm or less.

Watanabe teaches a monoclonal-based an immunochromatographic rapid assay for monensin (Watanabe, see title). Watanabe teaches a rapid test kit based on the immunochromatographic method to detect monensin in cattle milk, cattle plasma, and chicken plasma (Watanabe, page 2573, Abstract, lines 8-10 and Figure 4).

Matsuoka teaches that horses are much more sensitive to monensin toxicity than cattle (Matsuoka, page 8, ‘Summary’, lines 1-3). Matsuoka further teaches that monensin was introduced as a feed additive to promote increased feed efficiency in cattle (Matsuoka, page 8, ‘Introduction’, lines 7-9).

Matsuoka further teaches that horses eating supplement containing high levels of monensin results in toxicity and/or death (Matsuoka, page 8, 2nd paragraph, lines 1-5).

It would have been *prima facie* obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to have modified the device of Caussette, in order to have provided on the device, anti-monensin binding protein as the antibiotic binding protein located in the conjugate pad and further modify the test line with an immobilized binding reagent capable of binding the monensin binding protein, as an obvious matter of a simple substitution of one art recognized antibiotic binding protein (namely a monensin binding protein) and binding reagent (namely monensin) over another. The prior art contained the base invention (see as taught by Caussette, the prior art already recognized a rapid diagnostic test strip configured for a competitive assay for detecting antibiotics in a liquid sample, such as a liquid sample derived from animal feed).

Further, at the time the prior art recognized the ability of detecting monensin in a lateral flow assay (as in Watanabe). As such, it would have been obvious to modify the test strip of Caussette to capture monensin, as the ordinarily skilled artisan would appreciate one binding protein is usable in place of the other, given that Caussette teaches a lateral flow device for a competitive antibiotic detection assay and Watanabe teaches detection of monensin by a lateral flow device, and the artisan of ordinary skill would expect success either in a competitive (as taught by Caussette) or direct (as taught by Watanabe) assay, which are merely two different formats of a lateral flow assay, both doable on a single lateral flow device. The results would have been predictable, namely in that the device of Caussette in view of Watanabe would be expected capable of detecting monensin.

One having ordinary skill in the art would be motivated to have modified the device of Caussette (intended for detecting antibiotic in animal feed), specifically to detect monensin in animal feed because the art teaches that monensin was introduced as a feed additive to promote increased feed efficiency in cattle (Matsuoka, page 8, 'Introduction', lines 7-9), that horses eating supplement containing high levels

of monensin results in toxicity and/or death. As such, one of ordinary skill in the art would be motivated to modify the composition of Caussette in order to detect monensin to screen the feed to determine if it is feed that is also suitable for horses since horses are much more sensitive to monensin toxicity than cattle and that ingesting supplements high in monensin can lead to toxicity and/or death.

Berlina teaches a lateral flow immunoassay for the detection of antibiotics (Berlina, page 17, Abstract, line 3). Berlina further teaches that in a competitive lateral flow assay the concentration of the specific antibody is one of the most important parameters and that increasing antibody concentration over a concentration that gives an adequate analytical signal increases unspecific staining (Berlina, page 20, 2nd column, see entire 2nd paragraph).

Godoy-Navajas teaches a competitive heterogeneous immunoassay with antibody capture involving immobilization of BSA-monensin. Godoy Navajas further teaches adding 75 µl of 10 nM NP-labeled anti-monensin antibody (Godoy-Navajas, page 197, see '2.3.4. Determination of monensin'). Considering that an IgG antibody is defined by the state of the art as having a molecular weight of approximately 150 kDa (see e.g., Janeway, see 3-1., lines 1-2), Godoy-Navajas teaches adding 20 µg/ml labeled antibody.

Li et al. teaches a lateral flow device (see Li et al. CN108646036A, original, page 1 of 10). Li et al. further teaches a test strip to detect follicular stimulating hormone and interstitial cell stimulating hormone comprising a bonding pad comprising mouse anti-alpha follicular stimulating hormone monoclonal antibodies compromising colloid gold label (Li, CN108646036A, translation, from here on out referred to as Li, page 1 of 6, 'Abstract', lines 1-4). Li further teaches that the gold-labeled antibody crosses the nitrocellulose membrane (Li, page 4 of 6, paragraph 11, lines 1-2). Li further teaches a strip comprising 25-35 µg/ml gold labelled mouse anti-alpha-FSH monoclonal antibody on the bonding pad (Li, page 1 of 6, see paragraph '2.'). Li further teaches an embodiment of the test strip comprising 30 µg/ml gold labeled antibody (Li, page 4 of 6, 'Embodiment one', see entire 3rd paragraph). Put another

way, Li teaches 25-35 µg/ml gold labeled antibody loaded on the conjugate pad capable of binding a specific analyte (equivalent to the "M antibody").

Grice teaches a colloidal gold immunochromatographic test strip comprising detection lines, antigen lines and quality control lines and an anti-target protein monoclonal antibody-colloidal gold marker on the gold label pad for detecting gluten (Grice, page 1 of 6, 'Abstract', lines 1-8). Grice further teaches that the concentration of the anti-gluten protein monoclonal antibody-colloidal gold marker is 2-30 µg/ml (Grice, page 2 of 6, 'Contents of the invention', see 7th paragraph). Grice further teaches that when the sample is added to the sample pad it migrates to the gold label pad and if there is analyte to be detected it will combine with the gold-labeled antibody to form a complex and binds to the capture protein on the detection line (Grice, page 2 of 6, see entire last paragraph). As such, Grice also teaches a gold labeled antibody capable of binding an analyte in a sample wherein the bound analyte/antibody complex migrate to the detection line (equivalent to the "M antibody") and the antibody is at a concentration of 2-30 µg/ml.

Regarding the limitations 30-40 µg/ml (claim 1) and 30 µg/ml (claim 20) and 40 µg/ml (claim 21) as presently recited in the claims; it has been previously established by the courts that a *prima facie* case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). In the present case, 30-40 µg/ml, 30 µg/ml, and 40 µg/ml is an amount overlapping with or substantially close to what was taught in the prior art. Both Li and Grice teach a colloidal gold labeled antibody that binds an analyte in a sample and migrates along the lateral flow test strip, such as 25-35 µg/ml gold labelled antibody as taught by Li, 2-30 µg/ml gold labeled antibody as taught by Grice, or 20 µg/ml as taught by Godoy-Navajas. Berlina teaches that the concentration of the specific antibody is one of the most important parameters. Berlina further teaches providing enough antibody to give an adequate analytical signal and that increasing antibody concentration increases unspecific staining (Berlina, page 20, 2nd column, see entire 2nd paragraph). As

such it would have been considered *prima facie* obvious in view of the prior art to have arrived at the presently claimed antibody concentration in the course of routine experimentation, in order to achieve a desired effect of detecting monensin in a sample with an adequate signal, but without an increase in non-specific binding.

The courts have also held that a particular parameter, when recognized as a result effective variable, i.e., a variable that achieves a recognized result, may be subject to routine optimization in order to determine optimum or workable ranges of said variable (also MPEP 2144.05). In the present case, it would have been *prima facie* obvious to one of ordinary skill in the art to determine the optimum antibody concentration by optimizing what was known in the art to be a suitable concentration of antibodies as taught by Godoy-Navajas, Li, and Grice. Godoy-Navajas, Li, and Grice show that it was known in the art to use an antibody concentration of 2-35 µg/ml gold labeled mobile detection antibody for a lateral flow immunoassay.

It would have been *prima facie* obvious to have arrived at the claimed range between 30 and 40 µg/ml as an obvious matter of routine experimentation because the antibody concentration is an art recognized result effective variable (namely the variable which results in successful detection of monensin, that it would be desirable to optimize conditions to uncover the optimal concentration in order to avoid unnecessary increase in non-specific signal, see Berlina, concentration of the specific antibody is one of the most important parameters and comprising enough antibody to give an adequate signal and not so much that it increases unspecific staining). It would have further been obvious to have modified the 2-35 µg/ml in order to arrive at the claimed range/concentration values, one motivated to optimize the 2-35 µg/ml because of the teaching of Berlina that the concentration of the specific antibody is one of the most important parameters in a competitive lateral flow assay.

One of ordinary skill in the art would have a reasonable expectation of success because an antibody concentration of 2-35 µg/ml is substantially similar, and overlaps with the lower concentration

of the range 30-40 µg/ml, and considering the values are substantially close, there is no reason to expect that the values within the claimed range would achieve a different or unexpected result compared to 2-35 µg/ml. One of ordinary skill would expect success optimizing by routine experimentation because such optimization would allow the operator to achieve the desired detection of monensin and further because optimization is well within the skill level of the ordinary artisan, as supported by the above cited prior art.

Regarding the limitations that the device is a competitive lateral flow assay strip wherein the LFA strip device has a sensitivity of 10 ppm or less for detecting antibiotic in the liquid sample. As indicated in the citation of Caussette and the combined cited art above, the claimed invention is structurally indistinct from the cited prior art, and as such, would be expected capable of the limitations as recited.

Response to Arguments

12. Applicant's arguments filed 06/23/2025 have been fully considered but they are not persuasive.

Regarding the rejection of claims under 35 U.S.C. 103, Applicant argues, starting on page 7, that the invention is directed to a composition comprising a lateral flow assay to determine the presence of monensin in animal feed at high sensitivity in an aqueous sample rapidly and in a cost and labor saving manner. Applicant further argues that this was a long-felt need in the art at the time. Applicant argues (page 8) that the claimed invention is directed to compositions for carrying out a lateral flow assay to detect monensin and that the combination of features claimed are not taught or suggested by the cited references, particularly comprising the claimed aqueous extraction buffer and the device being loaded with 30-40 µg/ml of anti-monensin antibody, and further that this results in a sensitivity of 10 ppm or less for detecting monensin in the liquid sample in the aqueous detection buffer. Specifically, applicant argues that it was well established in the art that monensin is much more soluble in organic solvents

than in water (also shown in Example 1 of the specification). Applicant further argues that aqueous buffers have advantages, including ease of use and disposal, as well as being more compatible with lateral flow assays compared to organic solvents. Applicant argues that there was no reasonable expectation of success since there were no lateral flow assay devices or assays for detecting monensin in the art and thus it was unknown whether the degree of sensitivity needed for the claimed invention could be achieved.

This argument is not persuasive.

Watanabe demonstrates success using a monensin specific lateral flow assay, as such showing that monensin in a sample diluted in aqueous buffer can be detected by lateral flow assay. Further, Mount teaches diluting samples and monensin standards in phosphate buffered saline Polyethylene glycol sorbitan monolaurate in an immunoassay. Regarding the sensitivity of the assay, the combination of Caussette, Watanabe and the cited art above teaches the composition as claimed, including a lateral flow device comprising labeled detection antibodies at a concentration between 2-35 μg/ml. The prior art further supports that antibody concentration of the conjugate provided in a lateral flow device is a result effective variable and that it would be well within the skill of one of ordinary skill in the art to arrive at the claimed antibody concentration by routine optimization. As explained previously in detail above, because the combination of the prior art teaches the composition as claimed, the composition as taught by the combination of the prior art would be expected capable of determining the presence of monensin in animal feed at high sensitivity in an aqueous sample rapidly and in a cost and labor saving manner. Put another way, the combination of the prior art teaches a competitive lateral flow assay as claimed (cost and labor saving), comprising the anti-monensin specific antibody at the concentration as claimed (high sensitivity), and an aqueous extraction buffer consistent with those presently claimed, and as such the device/composition would be expected capable of the achieving the same sensitivity.

Regarding the argument that it is well established in the art that monensin is much more soluble in organic solvents than in water, this argument is not persuasive because Watanabe teaches a lateral flow assay for monensin comprising a sample, namely monensin, diluted in an aqueous buffer (TRIS-HCl).

Applicant further argues that there would be no reasonable expectation of success because Caussette and Berlina teach a lateral flow assay that does not disclose detection of monensin and that the Watanabe and Godoy-Navajas references each describe immunochromatographic assays that utilize an anti-monensin antibody, but are different types of assays and one of ordinary skill in the art would not have reasonably expected that results from one type of assay would translate or be relevant to another type of assay.

This argument is not persuasive.

Regarding the argument on page 9 that the lateral flow assay of Caussette and Watanabe are different and that the results from one assay would not be expected to translate to another type of flow assay, both assays require binding of an antibiotic to a specific antibody in an assay where a labeled binding partner is contacted in a flowing solution with the other binding partner and binds to such binding partner. Watanabe teaches an immobilized antibody that binds monensin in solution, whereas Caussette teaches immobilized antibiotic that is bound by specific antibody in solution (in the absence of the target in the sample). Put another way, Watanabe teaches success in detecting monensin in a liquid sample applied to a lateral flow assay comprising detection by binding to a specific antibody. Caussette teaches success detecting an antibiotic in an indirect immunoassay which relies on the binding of an antibiotic to its specific antibody in order to be detected in a competitive assay, and Watanabe teach a specific type of antibiotic, monensin.

Further, Godoy-Navajas teaches an assay with immobilized monensin to which a solution comprising monensin and anti-monensin specific antibody is added, teaching success in detecting

monensin in a competitive assay, the same assay principal as taught by Caussette. Put another way, Watanabe teaches binding of monensin to an anti-monensin antibody in a lateral flow assay and Godoy-Navajas teaches success using immobilized monensin to bind anti-monensin antibody in solution (a competitive assay). As such one of ordinary skill in the art would have a reasonable expectation of success because Godoy-Navajas teaches success using a competitive assay to detect monensin in solution and Caussette teaches success detecting an antibiotic in a competitive assay, where the antibiotic in solution is contacted with the antibody by lateral flow and Watanabe teaches success detecting monensin by lateral flow assay. Put another way, the prior art teaches success detecting monensin in a lateral flow assay and teaches success detecting monensin in a competitive assay, as well as teaching success detecting an antibiotic in a combination of the two, a competitive lateral flow assay and as such one of ordinary skill in the art would have a reasonable expectation of success detecting monensin in a competitive lateral flow assay.

Regarding the argument, also on page 9, that the assay of Godoy-Navajas does not require movement of the sample to come into contact with monensin, this argument is not persuasive because Watanabe teaches an assay that requires movement of the sample comprising monensin, namely a lateral flow assay for the detection of monensin which comprises the binding of monensin to monensin-specific antibody while the sample is moving. Absent evidence to the contrary one having ordinary skill would expect that binding between monensin and the specific antibody would occur whether it is the antibody or the monensin that is immobilized, especially because Godoy-Navajas teaches success in detecting monensin in a competitive immunoassay.

Applicant further argues on page 10 that the claimed lateral flow device differs structurally from the composition used in the immunoassay of Godoy-Navajas as well as the composition used in the ELISA-type assay taught by the Watanabe reference, because the site of monensin immobilization and sample application are the same (i.e. a microwell) and the sample does not have to flow across a

membrane to interact and because of these structural differences between the claimed composition and the compositions of Godoy-Navajas and Watanabe there would be no reasonable expectation of success.

This argument is not persuasive because Godoy-Navajas is not applied as anticipating the claimed invention, rather it is the combination of the cited art which addresses the claims. Watanabe teaches both an enzyme linked immunosorbent as well as a lateral flow assay (and teaches success in detecting monensin in both assay types) and further the lateral flow assay taught by Watanabe does not rely on a structure where the site of monensin immobilization and sample application are the same. Further, Godoy-Navajas is not relied on to teach the structure of the lateral flow assay but rather to teach that monensin, which is successfully detected by lateral flow assay (see Watanabe), can also be detected by an indirect immunoassay. Caussette teaches the competitive lateral flow assay device as claimed. Godoy-Navajas and Watanabe are relied on because they demonstrate the ability to perform detection of monensin by competitive assay (Godoy-Navajas) and by lateral flow assay (Watanabe). As discussed previously above, the artisan of ordinary skill would have a reasonable expectation of success using the competitive lateral flow assay of Caussette to detect monensin because Caussette demonstrates the ability to detect an antibiotic using competitive latera flow assay, and further the teaching of Godoy-Navajas supports even further that monensin can be detected using a competitive assay binding format.

Applicant further argues on page 10 that the antibody concentration taught by Godoy-Navajas is well outside the antibody concentration range specified in the claims and nothing in any of the cited references teaches or suggests the claimed concentration range.

This argument is not persuasive.

The prior art of Godoy-Navajas, Li, and Grice teach a range of labeled detection antibodies that overlaps with or is substantially close to the concentrations as claimed (see as previously explained in the rejection under 35 U.S.C. 103). As discussed previously above, it has been previously established by the courts that a *prima facie* case of obviousness exists where the claimed ranges or amounts do not overlap with the prior art but a merely close (see MPEP 2144.05). The amount of antibody as taught by Godoy-Navajas, Li, and Grice is close to or overlaps the amount as claimed, and the prior art (Berlina) supports that it is necessary to establish concentration that achieve a desired effect of detecting monensin in a sample with an adequate signal, but without an increase in non-specific binding (see Berlina above). As a result, it would have been obvious to have optimized, namely by selecting from within the range of prior art disclosed concentrations, to determine the optimum concentration for detection. The concentration of the antibody is considered a result effective variable, i.e. achieves the result of detecting without increasing background (referring to Berlina) and as such may be subject to routine optimization (also MPEP 2144.05). Therefore, even though the amount of antibody as taught by Godoy-Navajas, Li, and Grice is close, as in the case of Godoy-Navajas, or overlaps (see Li and Grice) but does not teach the exact range of the claimed antibody concentration, it is sufficiently close that the artisan of ordinary skill would have a reasonable expectation of arriving at the claimed concentration by routine optimization.

Applicant further argues that in view of the deficiencies in the teachings of the cited references as well as the lack of any LFA-type assay for monensin in the art and the low solubility of monensin in water, there was no reasonable expectation that the composition as set forth in the claims could successfully achieve a level of monensin detection sensitivity of 10 ppm or less. In particular, applicant argues, that applicant's discovery that a loading concentration of 30-40 μ/ml of anti-monensin M

antibody is able to achieve the desired level of sensitivity of 10 ppm or less using an aqueous extraction buffer.

This argument is not persuasive.

Watanabe teaches a monoclonal-based an immunochromatographic rapid assay (i.e. a lateral flow assay) for monensin (Watanabe, see title). Watanabe further teaches an aqueous assay buffer that is Tris-HCl and 0.05% Polyethylene glycol sorbitan monolaurate-20 (Watanabe, page 2574, 'Buffers', lines 1-3). Watanabe further teaches that the detection limits of monensin in cattle milk, cattle plasma, and chicken plasma were about 40-160 ppb, depending on the sample matrix (Watanabe, page 2573, 'Abstract', lines 10-11). Put another way, Watanabe teaches success in detecting monensin in a lateral flow assay at a higher sensitivity than claimed (Watanabe teaching ppb, the claimed method reciting ppm) while using a sample buffer that is aqueous, and as such it would not be unexpected that the composition as claimed (and as taught by the combination of the prior art) would be expected to successfully achieve the detection sensitivity as claimed. For the reasons discussed in the rejection in detail above, the composition as claimed is considered obvious over the combination of the prior art and in particular the artisan of ordinary skill would have arrived at the antibody concentration out of the course of routine experimentation. Therefore, the composition as taught by Caussette and the combined cited art above is structurally indistinct from the claimed composition, and as such, would be expected capable of the limitations as recited.

Applicant further argues secondary indicia of non-obviousness on page 10, including that the claimed invention met a long-felt need in the art and the claimed invention exhibits commercial success as evidenced by the accompanying declaration.

In the declaration, applicant recites (under 2.) that previously, only one commercial source of antibiotic monensin for use in animal feed included means for detecting the amount of monensin in

animal feed. The detection of monensin is facilitated by microtracers in the feed that are added along with the monensin. The detection of the microtracers therefore indicates the presence of monensin in animal feed.

According to applicant's arguments on page 11, this shows that at the time of the invention, the only commercial source of monensin for animal feed used a microtracer system.

Applicant further argues in the declaration (under 3.) that the microtracer system is described in Appendix A as small stainless-steel particles coated with food grade dye and added to monensin at low levels so that the presence or absence of monensin can be confirmed, though not as accurately as a laboratory assay.

According to applicant's arguments, page 11, the observation that detection of the microtracer particles in the feed provides only a ballpark estimate" of the amount of monensin in the feed further evidences the advantages of the claimed invention.

Applicant further argues in the declaration that feed mills reported that the monensin product as proposed by applicant would need to include a means for detecting monensin in the animal feed due to the potential toxicity of monensin at too high a level (see 4.). Applicant further recites an expressed need by farmers for a detection system that directly detects monensin itself to a detection threshold of 10 ppm on site using an aqueous sample (see 5.).

Applicant elaborates in arguments (page 11-12) that a monensin product would need to include means for detecting monensin in animal feed due to the potential toxicity of monensin if present at too high levels.

Applicant further argues in the declaration that given the low solubility of monensin in water there was no reasonable expectation that one of skill in the art could achieve the claimed invention for detecting monensin at a sensitivity threshold of 10 ppm (see 6.).

In the declaration (under 7.), applicant recites that sales of the Monovet system have increased year-over-year since the product launched in 2019 and that it has captured approximately 1/3 of the US monensin market (see Appendix C), which according to applicant's arguments on page 12, demonstrates the ongoing commercial success of Monovet with the companion Quicktest (the lateral flow device as claimed).

Applicant recites under 8. that prices for both the system comprising the lateral flow assay system and the system comprising the microtracers are equivalent, which according to applicant's opinion is due to the companion Quicktest monensin detection system, since this direct detection system has numerous advantages (see 9.). According to applicant's arguments (page 12) this provides evidence that the commercial success of the monensin system is due to its advantageous combination with the quicktest.

Applicant also recites in the declaration (under 10.) and the arguments on page 13 that there is also demand from some customers to purchase the quicktest system itself to detect monensin in other products, which indicates that the claimed invention is meeting a long felt unmet need in the art.

According to applicant's arguments (page 13), for the reasons discussed in the arguments and the accompanying declaration, Applicant asserts that the marketplace response to the monensin quicktest provides the necessary evidence and nexus to establish that the claimed invention meets a long felt unmet need in the art and exhibits commercial success and therefore claims 1-4, 6-9, and 20-21 are nonobvious over the cited references.

These arguments are not persuasive.

Applicant's arguments suggest that prior to the invention, no one in the prior art took the step towards developing a direct assay for the detection of monensin in a liquid sample. However, prior to Applicant's filing date, Watanabe took that step and taught a lateral flow device for the detection of monensin in a liquid sample. Applicant argues that the observation that their monensin product

comprising a test device as claimed has steadily gained market shares since it was brought to market and the fact that some customers have asked for the test device separately shows that there was a long unmet need for the direct detection of monensin. However, there is no evidence showing that the features distinguishing the lateral flow test as claimed from the indirect detection method for monensin in other monensin products is responsible for the commercial success of the claimed device and further there is no evidence for a recognized, persistent unmet need.

Further, as discussed previously above, the problem of a direct detection device for the detection of monensin in a liquid sample was solved by the teaching of Caussette. Caussette is teaching the assay format, a competitive lateral flow assay for the detection of an antibiotic in samples such as feed. It was known in the art at the time that monensin as a feed additive promotes increased feed efficiency in cattle but also that horses eating supplement containing high levels of monensin results in toxicity and/or death (see Matsuoka) and as such one of ordinary skill in the art would have been motivated to apply the competitive lateral flow test device as described by Caussette to detect monensin in a liquid animal feed sample. Monensin is a specific type of antibiotic similar to what is taught by Caussette and Watanabe teaches success in detecting monensin in a lateral flow assay. In doing so, Caussette in view of Watanabe supplied the key element required to satisfy the long-felt need that Applicants contend was recognized in the art. *Newell,* 864 F.2d at 768 ("[O]nce another supplied the key element, there was no long-felt need or, indeed, a problem to be solved."). The composition set forth in claim 1 is obvious over the combined teachings of Caussette and the prior art above. Despite the observation that the claimed device, has the advantage over microtracer assays of directly and accurately detecting monensin in animal feed without the need for microtracers, both systems are capable of detecting monensin in animal feed with sufficient sensitivity to avoid toxicity. The need of testing animal feed to ensure its safety for susceptible animal species is met by both the claimed device

and the microtracer system and as such does not appear to be sufficient evidence of an unmet need in the art.

Regarding the argument that the detection of monensin in aqueous extraction buffer is an unexpected result, both Watanabe and Godoy-Navajas teach aqueous buffers for the detection of monensin and Mount teaches an immunoassay detecting monensin in phosphate buffered saline comprising 0.5% Polyethylene glycol sorbitan monolaurate-20. Further, the combination of the prior art teaches a composition as claimed, and as such, the device as taught by the combination of the prior art would be expected capable of detecting monensin in aqueous extraction buffer at the sensitivity as claimed.

Applicant further argues that the rejection of claims 5 and 22 should be withdrawn for the same reasons as discussed for claims 1-4, 6-9, and 20-21. However, for the reasons above the rejection of claims 1-4, 6-9, and 20-21 is maintained and therefore the rejection of claims 5 and 22 is also maintained.

For all the reasons above the arguments are not persuasive.

Communication

13. Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEFANIE J KIRWIN whose telephone number is (571)272-6574. The examiner can normally be reached Monday - Friday 7.30 - 4 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached at (571) 272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/STEFANIE J. KIRWIN/
Examiner, Art Unit 1677

/ELLEN J MARCSISIN/
Primary Examiner, Art Unit 1677